



Rugby River Champion's Risk Assessment



UNIVERSITY OF
BIRMINGHAM

General Health and Safety

Volunteer safety is incredibly important when carrying out any surveying as a River Champion. The following advice should always be followed when surveying:

- You must complete training through trained personnel from the University of Rugby before carrying out ANY sampling with Rugby River Champions.
- Always carry out sampling and surveying with another person, do not enter a river or body of water alone.
- Wear waterproof, closed toed shoes with suitable grip when carrying out any activities. ideally wellies.
- Only access water that is below “welly height” (around 40cm).
- Have a mobile phone on your person to call for help (dial 999 if necessary).

The following table highlights specific hazards you may encounter whilst surveying, as well as additional measures to reduce risk for yourself and the people around you.

Activity	Hazard	Risk	Control Measures
Entering the water for Urban Riverfly or Water Chemistry	Slips, trips and falls	Hazard may occur whilst entering/ existing water, or whilst kick sampling.	• Check that riverbed sediments underfoot are stable using the kick net base (or a staff / wading pole). • Beware of and avoid ‘slimy’ substrate (e.g., rocks covered in wet algae / mosses).
	Cuts / puncture wounds	Sharp objects may be present at sampling sight Murky water may reduce riverbed visibility Floating debris may impact a volunteer.	• Visually check area for sharps before beginning survey • Take extra care when entering murky water, do not enter if you do not feel safe doing so. • If equipment is damaged, do not continue to use it.

Activity	Hazard	Risk	Control Measures
Entering the water for Urban Riverfly or Water Chemistry	Foot entrapment	Hazard may occur whilst entering/ existing water, or whilst kick sampling, particularly in murky water.	• Check that riverbed sediments underfoot are stable using the kick net base (or a staff / wading pole). • Beware of and avoid 'slimy' substrate (e.g., rocks covered in wet algae / mosses).
	Immersion, drowning	May be caused by slips trips and falls, deep waters, fast flows and an uneven riverbed, unexpected sudden increase in flow, etc.	• Check the riverbed depths and type of sediment using the kick net base (or a staff / wading pole). • Check the weather forecast and river levels (if applicable) and abandon monitoring if heavy rainfall or flood conditions are occurring or expected. • Abandon monitoring if water levels begin rising rapidly and become notably more murky. • Wear appropriate, waterproof clothing based on weather conditions. • You may wish to wear a buoyancy aid during sampling. Buoyancy aids must be provided by the individual/volunteer group.
	Exposure to disease and infection	Volunteers are at a higher risk if they have open wounds, are bitten, or accidentally come into contact with water contaminated with urine from infected rats. Infection can enter the body through cuts and scratches, eyes and the mouth.	• Do not enter the water if you have cuts or broken skin. If you take the risk of entering the river, consider covering all cuts and broken skin with waterproof plasters. • Wash hands before and after contact with water and prior to eating, drinking, smoking.
	Risk to the environment	Disturbance of silt and mud (downstream pollution) Spread of infection to / from other water courses Spread of invasive species to / from other sites Distress to fish and other wildlife e.g. disturbance of fish spawning sites and nesting birds	• Disinfect after entering water courses • Inform volunteers of risk during training, sharing reminders in monthly newsletter when appropriate • Follow government guidelines for managing invasive species should one be encountered.

Activity	Hazard	Risk	Control Measures
Surveying outdoors	Bites and stings	Bites / stings / irritants from nature / allergies	• Be aware of and avoid potentially harmful flora—particularly giant hogweed along riverbanks (its sap can cause nasty burns), and any other toxic plants. • Be aware of and where possible avoid potentially harmful fauna like biting midges, bees and wasps. • Wear insect repellent and carry appropriate treatment where necessary (e.g., antihistamines, epinephrine / adrenaline autoinjectors for those with allergies). Individual volunteers are responsible for managing their own medication. • Wear long and relatively tight (i.e., not loose) clothing. • Be aware of the symptoms of Lymes disease, contact your general practitioner if you are showing symptoms.
	Weather	Medical issues from hot and dry weather (e.g., dehydration, sun stroke, sunburn). Medical issues from wet and cold weather conditions (e.g., hypothermia). Safety issues when stormy (wind and lightening).	• Check weather forecast and wear appropriate clothing accordingly (taking a spare set of clothing in case of any accidents). • Take sun cream (SPF 15+). Volunteers are responsible for providing their own suncream. • Drink plenty of fluids, particularly in warm weather. • Do not survey in lightning conditions. • Do cautious of surveying near trees during windy weather. Do not survey if wind speeds exceed 32 miles per hour (<u>near gale speeds</u>).
	Members of the public	Other activities Threatening behaviours	• Be aware of people outside of your volunteer group in the vicinity • If you feel unsafe, abort surveying and leave the area • Dial 999 if required
	Traffic	Vehicular movements	• Practice care and attention before crossing roads

Activity	Hazard	Risk	Control Measures
Surveying outdoors	Livestock and agriculture	Agricultural practices may be taking place near by, including grazing livestock, chemical use, etc. Fencing may also be present	- Do not access a site without permission from the landowner - Be aware of, and keep well away from, nearby livestock. Always be aware of your nearest exit. - Access sites using safe means, do not climb over barbed wire. - If site is actively being sprayed, do not enter. Return another day.
Kick sampling Chemical testing	Chemical Use	The Freshwater Watch kits measuring nitrate and phosphate tubes contain a strong acid substance, while the reagents in the Hanna Ammonia checker are highly toxic and corrosive. Both are contained securely within appropriate containers and the techniques allow the solutions to be used very easily.	- Do not allow any individual who has not participated in the relevant training to use the product and store out of reach from children / animals. - Wear the eye goggles and gloves provided when sampling and take extra caution to ensure these toxins remain within the parameters of the respective sampling procedure. - Dispose of the sample biproducts down a sink. For the nitrate / phosphate tubes, squeeze the solution out in its entirety and then recycle the drained tube. For the ammonia, empty the vial and then rinse with tap water. Rinse and wipe down the sink surface after the solutions have been disposed of. - If any contact occurs with eyes and skin, or enters the mouth, rinse and wash the areas thoroughly with plenty of clean water. Seek medical advice immediately if an adverse reaction occurs or if any substances are ingested. - Read the supplier's safety data sheets for the ammonia (Hanna instruments), nitrate and phosphate test kits for further information.

Activity	Hazard	Risk	Control Measures
Kick sampling Chemical testing	Muscular Skeletal Disorder	Repetitive kick sampling. Manual handling / lifting. Posture.	- Awareness of posture - Correct lifting techniques and back care. If you do not feel that you can lift something, do not lift it.
General	Personal health	Lack of individual capability. Poor fitness Pre-existing injuries.	- Only participate if you feel you are able to - Reading, understanding and implementing the health and safety information reported in this document - Pre-existing injuries can be exacerbated by the activity. If the effects are felt post monitoring, seek quick medical advice / treatment (delay can significantly hinder recovery)

This document aligns with the General Risk Assessment submitted to the department of Geography, Earth and Environmental Services at the university of Birmingham, reference RA0002114.

Should you have any questions about this document, or about becoming a Rugby River Champion, please contact us via rugbyriverchampions@gmail.com



@RugbyRiverChampions

rugbyriverchampions@gmail.com



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Chemical Hazard and Risk Assessment

You will need the Safety Data Sheet(s) for the substance(s) to fill out this form.

Data Sheets are available from the supplier or from ChemWatch.

Refer to Health and Safety Guidance – Chemical Hazard and Risk Assessment GUIDANCE/22/CHRA/05.



Name of Assessor:	Caroline Roper		
Name of Supervisor: If applicable			
School/Department:	Geography, Earth and Environmental Sciences		
Date of assessment:	18/09/2026	Assessment number:	

The Activity

Location of activity: Building and room number	Monitoring to be undertaken outdoors
Activity assessed:	Freshwater Watch nitrate citizen science measurements using Kyoritsu pack tests.

Intended use:

Give details of the procedure. You may attach a protocol / instructions. It may be useful to break the procedure down into stages (suggested maximum of 4), to facilitate consideration of risks and controls at each stage.

Stage 1	Take pin out of the top of the sealed plastic tubes containing reagents, press in and suck sample water in
Stage 2	
Stage 3	
Stage 4	

Justification:

Justification is needed for exceptionally hazardous substances. These include carcinogens, mutagen, teratogens, asthmagens and highly spontaneous reactive substances

This technique affords a cost-friendly and accurate way of measuring ammonia, which can detect sewage pollution. The toxic chemicals are well contained and the sampling will be done in the outdoors (well ventilated).

Frequency of the activity:	Monthly	Duration:	5 minutes
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Materials Involved

Include all materials used, produced and/or encountered.

Name	CAS no.	Quantity*	Form	Hazards	H numbers	H statements	WEL**	Toxicity rating***
PACKTEST Nitrate (containing Zinc powder, Polyethylene and N, N-Diethy-N'-(1-naphthyl) ethylenediamine oxalate)	7440-66-6 / 9002-88-4/ 29473-53-8	140g	Powder	Warning	H317	May cause an allergic skin reaction.		

*Quantity should be the largest amount handled

**WEL Workplace Exposure Limit (see [GUIDANCE/22/CHRA/05](#) for further information)

***Toxicity rating. Rated on a scale of 1 to 4 using the CLP Scale, given in Appendix 1 (see [GUIDANCE/22/CHRA/05](#) for further information)

Reportable substances

Do any materials involved contain the following?

H317, H334, H340, H350, H350i, nanomaterials, metal working fluids, used engine oils or respirable crystalline silica.

☒ YES

☐ NO

If you have answered 'Yes' to the above, please complete or update a Health Record: https://universityofbirmingham.service-now.com/hr_core?id=manage_health_records

☒ HEALTH RECORD
COMPLETED

Risks

Specify risks to H&S from intended use for each stage identified above:

For procedures consisting of more than 4 stages, please continue in Appendix A.

	Possible route(s) of entry <i>(Select all that apply at each stage)</i>	Process factors influencing the risk of exposure <i>(Select all that apply at each stage)</i>	Risk implications <i>(Select all that apply for the procedure as a whole)</i>	Additional comments on the hazards associated with these substances
Stage 1	☒ In contact with skin and eyes ☐ Inhalation ☒ Ingestion ☒ Direct entry <i>e.g. open wounds, injection</i> ☐ Other <i>(please specify)</i> :	☐ Weighing ☐ Spraying ☒ Pouring ☐ Addition of solid reagents ☐ Pipetting ☐ Filtering ☐ Shaking/mixing ☐ Centrifugation ☐ Use of sharps ☐ Elevated temperature ☐ High pressure ☐ Sonication ☐ Grinding ☐ Cutting ☐ Welding ☐ Other <i>(please specify)</i> :	☐ Fire or explosion risk? <i>If yes, include prevention.</i> <i>(Include flash points, auto-ignition temperature and/or any other relevant information.)</i>	
Stage 2	☒ In contact with skin and eyes ☐ Inhalation ☒ Ingestion ☒ Direct entry <i>e.g. open wounds, injection</i> ☐ Other <i>(please specify)</i> :	☐ Weighing ☐ Spraying ☒ Pouring ☐ Addition of solid reagents ☐ Pipetting ☐ Filtering ☐ Shaking/mixing ☐ Centrifugation ☐ Use of sharps ☐ Elevated temperature ☐ High pressure ☐ Sonication ☐ Grinding ☐ Cutting ☐ Welding ☐ Other <i>(please specify)</i> :	☐ Chemical reactions or thermal runaway? <i>If yes, include containment/ incompatibilities with other substances, hazardous decomposition products.</i>	
Stage 3	☐ In contact with skin and eyes ☐ Inhalation ☐ Ingestion ☐ Direct entry <i>e.g. open wounds, injection</i> ☐ Other <i>(please specify)</i> :	☐ Weighing ☐ Spraying ☐ Pouring ☐ Addition of solid reagents ☐ Pipetting ☐ Filtering ☐ Shaking/mixing ☐ Centrifugation ☐ Use of sharps ☐ Elevated temperature ☐ High pressure ☐ Sonication ☐ Grinding ☐ Cutting ☐ Welding ☐ Other <i>(please specify)</i> :	☐ Gas release or oxygen deficiency? <i>If yes, include control measures.</i>	
Stage 4	☐ In contact with skin and eyes ☐ Inhalation ☐ Ingestion ☐ Direct entry <i>e.g. open wounds, injection</i> ☐ Other <i>(please specify)</i> :	☐ Weighing ☐ Spraying ☐ Pouring ☐ Addition of solid reagents ☐ Pipetting ☐ Filtering ☐ Shaking/mixing ☐ Centrifugation ☐ Use of sharps ☐ Elevated temperature ☐ High pressure ☐ Sonication ☐ Grinding ☐ Cutting ☐ Welding ☐ Other <i>(please specify)</i> :	☐ Other <i>(please specify)</i> :	

Persons who may be at risk:
Who is exposed to it?

- ☒ Departmental Staff
- ☐ Undergraduate students
- ☒ Postgraduate students
- ☐ Visitors
- ☐ Estates / cleaning / security staff
- ☐ Contractors, including service engineers
- ☐ Young workers
- ☐ Immuno-compromised individuals
- ☐ Pregnant women or women of reproductive age

☒ Others. Please specify:
(Include names if appropriate)

Volunteers that have signed up to the Birmingham River Champions project. They've had to confirm they've read a risk assessment (F-RAMP approved) within a sign up process

Notification method:
Groups at risk of exposure should be informed of any risk to which they may be exposed from this work activity

- ☐ Warning notice displayed on door
- ☐ Warning notice displayed within work area (e.g. laboratory bench, fume cupboard etc.)
- ☐ Hazards will be discussed with other workers
- ☐ Email notification to be sent to other workers

☐ Other. Please specify:

If nanomaterials are being used, please categorise the nanomaterial risk-level:

☐ HIGH:	Dry, dispersible nanomaterials or nanomaterial agglomerates / aggregates
☐ MEDIUM:	Nanomaterials suspended in liquids
☐ MEDIUM:	Nanomaterials attached to a 2D surface
☐ LOW:	Nanoparticles embedded into a solid 3D matrix

Approval for low- and medium-risk work involving nanomaterials is via principal investigator. Approval for high-risk nanomaterials work granted by either by the Safety Committee or jointly by the Safety Committee and the H&S Coordinator.

Control Measures

Could the use of hazardous substances be eliminated?
(If so, you should do so)

☐ YES ☒ NO

Could you substitute any of these materials for less hazardous alternatives or forms?
(If so, you should do so)

☐ YES ☒ NO

Controls and PPE Section 7, 8 & 15 of the Safety Data Sheet

What controls are required for these substances?

For procedures consisting of more than 4 stages, please continue in Appendix A.

	Engineering controls	Special requirements (e.g. special glassware; no vibration; Fluoro-plastic apparatus; no dry chemicals on heated surfaces; must be in the dark, not in spray/mist form, reduction in number of people exposed, authorised persons only)	PPE (PPE is a last resort and should be used in addition to all other control measures identified if the combination of these controls fails to achieve adequate control of exposure)
Stage 1	☒ None required ☐ A recirculating fume cupboard ☐ A ducted fume cupboard ☐ A glove box ☐ Local Exhaust Ventilation ☐ Other / additional control measures. Please specify: (e.g. a purpose-built facility, other specialist enclosure, well ventilated areas)	Eye protection and Polyethylene (PE) Gloves to be worn by all trained personnel undertaking the monitoring	☒ Eye protection? (State type required: e.g. safety spectacles, goggles, face shield) ☒ Gloves? (State type and thickness required, considering breakthrough times, permeation rates) ☐ Mask/RPE? (State type of mask/ filter) ☐ Laboratory coat or other clothing? (State type required) ☐ Other (please specify):

Stage 2	☒ None required ☐ A recirculating fume cupboard ☐ A ducted fume cupboard ☐ A glove box ☐ Local Exhaust Ventilation ☐ Other / additional control measures. Please specify: (e.g. a purpose-built facility, other specialist enclosure, well ventilated areas)	Eye protection and Polyethylene (PE) Gloves to be worn by all trained personnel undertaking the monitoring	☒ Eye protection? <i>(State type required: e.g. safety spectacles, goggles, face shield)</i> ☒ Gloves? <i>(State type and thickness required, considering breakthrough times, permeation rates)</i> ☐ Mask/RPE? <i>(State type of mask/ filter)</i> ☐ Laboratory coat or other clothing? <i>(State type required)</i> ☐ Other (please specify):
Stage 3	☐ None required ☐ A recirculating fume cupboard ☐ A ducted fume cupboard ☐ A glove box ☐ Local Exhaust Ventilation ☐ Other / additional control measures. Please specify: (e.g. a purpose-built facility, other specialist enclosure, well ventilated areas)		☐ Eye protection? <i>(State type required: e.g. safety spectacles, goggles, face shield)</i> ☐ Gloves? <i>(State type and thickness required, considering breakthrough times, permeation rates)</i> ☐ Mask/RPE? <i>(State type of mask/ filter)</i> ☐ Laboratory coat or other clothing? <i>(State type required)</i> ☐ Other (please specify):
Stage 4	☐ None required ☐ A recirculating fume cupboard ☐ A ducted fume cupboard ☐ A glove box ☐ Local Exhaust Ventilation ☐ Other / additional control measures. Please specify: (e.g. a purpose-built facility, other specialist enclosure, well ventilated areas)		☐ Eye protection? <i>(State type required: e.g. safety spectacles, goggles, face shield)</i> ☐ Gloves? <i>(State type and thickness required, considering breakthrough times, permeation rates)</i> ☐ Mask/RPE? <i>(State type of mask/ filter)</i> ☐ Laboratory coat or other clothing? <i>(State type required)</i> ☐ Other (please specify):

Monitoring Performance of control measures:

e.g. fume hood flow rates required, maintenance, examination and testing of engineering controls, PPE, RPE

☐ Fume cupboards and LEV:

- A thorough examination and test in the last 14 months, in accordance with University Policy and COSHH Regulations.
- A regular, routine maintenance schedule, in accordance with University Policy.
- User checks undertaken prior to each use, in accordance with University Policy.

☒ PPE/RPE:

- Users should meet the requirements for PPE and RPE, as set out in University Policy, including maintenance, inspection, and testing, and suitability assessment.
- Before each period of work, users must carry out an external visual inspection of all PPE to confirm suitability for the work activity and that it is not damaged or degraded so as to impair fit or performance.
- Respirators in regular use (other than one-shift disposable) must be thoroughly examined and, where appropriate, tested at least once a month or more frequently according to circumstances. Respirators used only occasionally should be examined and tested prior to next use and maintenance carried out as appropriate. N.B. The selection of a tight fit respirator for an individual wearer must include an appropriate fit test.

☐ Pressure vessels (e.g. liquid nitrogen transport dewars):

- Regular examination, inspection and testing, in accordance with University Policy and PSSR Regulations, and a written scheme is in place where required.
- Pressure vessels are checked regularly for general condition and leakage.

☐ Other measures not listed above (e.g. anemometers fitted to LEV equipment). Please specify:

Monitoring Workplace Exposure

Will monitoring for airborne contaminants or personal monitoring be required?

☐ YES

☒ NO

If yes, give details:

Monitoring Health Surveillance

Is there a known health impact from this level of exposure?

☐ YES

☒ NO

If yes, is health surveillance required to assess whether this health impact is developing in vulnerable individuals or all people exposed?

☐ YES

☒ NO

If yes, please raise a request with Occupational Health for Health Surveillance, via your line manager/PI by creating a new pre-placement at <https://birminghamuni.opasg2.com/>

Instruction and Training

Are any Standard Operating Procedures (SOPs) referred to in this protocol?

☐ YES

☒ NO

SOP reference number(s):

Specify training courses and/or special arrangements:

Storage and Waste Disposal

Storage

How should the materials be stored?

e.g. locked cupboard appropriately labelled and signed; stored away from other substances

Stored inside container away from non-trained individuals and wildlife.

Appropriate storage provision should be made prior to chemical purchase

Segregation *Section 10 of the Safety Data Sheet*

Is there any other chemical that these substances must not come in to contact with?

Procedure in case of reaction?

N/A – there are no particular risks of reactions with other substances under normal conditions

Disposal *Section 13 of the Safety Data Sheet*

How should the substance be disposed of?

Select all those that apply

☐ **Non-hazardous** *(Disposal via standard waste stream e.g. laboratory bin or sink)*

☐ **Halogenated solvent** - *an organic solvent containing halogens, i.e., Cl, F, Br, I (Disposal via specialist contractor through your local Health and Safety co-ordinator)*

☐ **Non-halogenated solvent** *(Disposal via specialist contractor through your local Health and Safety co-ordinator)*

☒ **Specialist hazardous waste disposal** - *including hazardous aqueous waste (Disposal via specialist contractor through your local Health and Safety co-ordinator)*

☐ **Treatment**

e.g. neutralisation of hazardous waste

Please detail:

Additional disposal details:

(Include 6-digit code and HP number if disposal is via external contractor and the coding is not undertaken centrally by your local H&S co-ordinator. See [Health and Safety Guidance - Hazardous Waste: Guidance on Assessment](#).)

Excess or expired reagent should be disposed of according to local regulations via the laboratory technician or manager.

Other Precautions and Emergency Procedures

Spillages or uncontrolled release *Section 6 of the Safety Data Sheet*

How should an accidental release or spillage be dealt with?

Consider both small- and large-scale spillages and release through failure of control measures

To control the hazards:

e.g. spread absorbent material on liquid spills, location of absorbent material, eliminate sources of ignition, isolation from power supply etc.

Prevent any further spillage if safe to do so.

The product must not come into contact with surface or ground water or enter the sewage system.

Put the leaked product into a suitable secondary container.

Dispose of all waste as hazardous materials.

To protect personnel:

Protection for personnel involved in the clean-up

Eye protection and protective gloves to be worn throughout the full procedure.

To render site of emergency safe:

Equipment needed for decontamination, clean-up etc.

Plentiful supply of clean water.

Evacuation required? ☐

First Aid *Section 4 of the Safety Data Sheet*

What should be done in the case of:

Ingestion:

Drink as much water as possible. Seek medical advice/attention immediately. Do not induce vomiting unless explicitly authorised by a doctor.

Eye contact:

Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Seek medical advice / attention immediately.

Inhalation:

Ensure surveyors are outdoors / getting fresh air away from the accident scene. First aid trained surveyors could administer artificial respiration. Seek medical advice / attention immediately if symptoms appear.

Skin contact:

Take off immediately all contaminated clothing. Rinse skin with water. Seek medical advice / attention immediately if symptoms appear

In the event of contamination or exposure, it is important to be able to contact a first aider without delay. Ensure you are aware of your local first aiders prior to commencement of work.

Fire Precautions *Section 5 of the Safety Data Sheet*

What actions should be taken in the event of fires involving these substances?

Avoid shock and friction. Avoid the bottle overheating

The first step should be to raise the alarm BUT if a fire extinguisher is needed, what extinguisher(s) should be used?

e.g. CO₂, foam, fire blanket, Class D Powder extinguisher

Any conventional fire extinguisher: carbon dioxide, foam, powder and water spray

Only tackle a fire if you have received training and it is safe to do so. Never put yourself at risk.

Emergency contact:

Name(s) and contact telephone number(s)

Review

Review date:

Maximum interval 3 years. Please note, some areas (including Chemistry) set a maximum review date of 1 year. Please check locally if unsure.

18/09/2029

You should review your risk assessment if you think it may no longer be valid (e.g. following significant changes to hazards, such as a change in process, materials, quantities, or following an accident, incident or ill-health.)

Assessment of Risk

Is the level of risk acceptable (are hazards to health adequately controlled with control measures in place)?

☒ YES

☐ NO

Overall assessment of risk:

Taking into account the above factors, what level of risk does the procedure pose?

- ☐ EXTREME: likely to present significant or permanent health effects to the majority of individuals
- ☐ HIGH: likely to present a significant risk to health to the majority of individuals
- ☐ MEDIUM: may present a significant risk to health to a minority of individuals
- ☒ LOW: the majority of individuals will be unaffected with only a minimal risk to health
- ☐ INSIGNIFICANT: no foreseeable risk of injury

N.B. Where risks associated with dangerous substances and/or explosive atmospheres are not adequately controlled, a DSEAR assessment should be undertaken to identify and assess the risks and determine any additional control measures required.

Declaration

Note: If working with nanomaterials, approval for low- and medium-risk work is via the Principal Investigator. Approval for high-risk nanomaterials work is to be granted by either by the Safety Committee or jointly by the Safety Committee and the Health and Safety Coordinator.

Assessor(s) Name:	Assessor(s) Signature:	Date:
Caroline Roper	Caroline Roper	18/09/2026

The line manager, supervisor or principal investigator should sign below to show that the assessment is a correct and reasonable reflection of the hazards and of the control measures and actions required:

Supervisor(s) Name:	Supervisor(s) Signature:	Date:

Appendix

Appendix 1 CLP Toxicity rating

Toxicity Rating	H number and statement
1/2	H300 Fatal if swallowed H310 Fatal in contact with skin H330 Fatal if inhaled

3	H301 Toxic if swallowed H311 Toxic in contact with skin H331 Toxic if inhaled
3 (ASPT1)	H304 May be fatal if swallowed and enters airways
4	H302 Harmful if swallowed H312 Harmful in contact with skin H332 Harmful if inhaled
Not classified as toxic	

University of Birmingham

Chemical Hazard and Risk Assessment

You will need the Safety Data Sheet(s) for the substance(s) to fill out this form.

Data Sheets are available from the supplier or from ChemWatch.

Refer to Health and Safety Guidance – Chemical Hazard and Risk Assessment GUIDANCE/22/CHRA/05.



Name of Assessor:

Caroline Roper

Name of Supervisor:

If applicable

School/Department:

Geography, Earth and Environmental Sciences

Date of assessment:

18/09/2026

Assessment
number:

The Activity

Location of activity:

Building and room number

Monitoring to be undertaken outdoors

Activity assessed:

Freshwater Watch phosphate citizen science measurements using Kyoritsu packtests.

Intended use:

Give details of the procedure. You may attach a protocol / instructions. It may be useful to break the procedure down into stages (suggested maximum of 4), to facilitate consideration of risks and controls at each stage.

Stage 1	Take pin out of the top of the sealed plastic tubes containing reagents, press in and suck sample water in
Stage 2	
Stage 3	
Stage 4	

Justification:

Justification is needed for exceptionally hazardous substances. These include carcinogens, mutagen, teratogens, asthmagens and highly spontaneous reactive substances

This technique affords a cost-friendly and accurate way of measuring ammonia, which can detect sewage pollution. The toxic chemicals are well contained and the sampling will be done in the outdoors (well ventilated).

Frequency of the activity:

Monthly

Duration:

5 minutes

Materials Involved

Include all materials used, produced and/or encountered.

Name	CAS no.	Quantity*	Form	Hazards	H numbers	H statements	WEL**	Toxicity rating***
PACKTEST Phosphate (containing Inosine, Polyethylene and 4-Aminoantipyrine)	58-63-9 / 9002-88-4 / 83-07-8	140g	Powder	None	-	-	-	-

*Quantity should be the largest amount handled

**WEL Workplace Exposure Limit (see [GUIDANCE/22/CHRA/05](#) for further information)

***Toxicity rating. Rated on a scale of 1 to 4 using the CLP Scale, given in Appendix 1 (see [GUIDANCE/22/CHRA/05](#) for further information)

Reportable substances

Do any materials involved contain the following?

H317, H334, H340, H350, H350i, nanomaterials, metal working fluids, used engine oils or respirable crystalline silica.

☒ YES

☒ NO

If you have answered 'Yes' to the above, please complete or update a Health Record: https://universityofbirmingham.service-now.com/hr_core?id=manage_health_records

☐ HEALTH RECORD COMPLETED

Risks

Specify risks to H&S from intended use for each stage identified above:

For procedures consisting of more than 4 stages, please continue in Appendix A.

	Possible route(s) of entry <i>(Select all that apply at each stage)</i>	Process factors influencing the risk of exposure <i>(Select all that apply at each stage)</i>	Risk implications <i>(Select all that apply for the procedure as a whole)</i>	Additional comments on the hazards associated with these substances
Stage 1	☒ In contact with skin and eyes ☒ Inhalation ☒ Ingestion ☒ Direct entry <i>e.g. open wounds, injection</i> ☐ Other <i>(please specify)</i> :	☐ Weighing ☐ Spraying ☐ Pouring ☒ Addition of solid reagents ☐ Pipetting ☐ Filtering ☐ Shaking/mixing ☐ Centrifugation ☐ Use of sharps ☐ Elevated temperature ☐ High pressure ☐ Sonication ☐ Grinding ☐ Cutting ☐ Welding ☐ Other <i>(please specify)</i> :	☐ Fire or explosion risk? <i>If yes, include prevention.</i> <i>(Include flash points, auto-ignition temperature and/or any other relevant information.)</i>	
Stage 2	☒ In contact with skin and eyes ☒ Inhalation ☒ Ingestion ☒ Direct entry <i>e.g. open wounds, injection</i> ☐ Other <i>(please specify)</i> :	☐ Weighing ☐ Spraying ☒ Pouring ☐ Addition of solid reagents ☐ Pipetting ☐ Filtering ☐ Shaking/mixing ☐ Centrifugation ☐ Use of sharps ☐ Elevated temperature ☐ High pressure ☐ Sonication ☐ Grinding ☐ Cutting ☐ Welding ☐ Other <i>(please specify)</i> :	☐ Chemical reactions or thermal runaway? <i>If yes, include containment/ incompatibilities with other substances, hazardous decomposition products.</i>	
Stage 3	☐ In contact with skin and eyes ☐ Inhalation ☐ Ingestion ☐ Direct entry <i>e.g. open wounds, injection</i> ☐ Other <i>(please specify)</i> :	☐ Weighing ☐ Spraying ☐ Pouring ☐ Addition of solid reagents ☐ Pipetting ☐ Filtering ☐ Shaking/mixing ☐ Centrifugation ☐ Use of sharps ☐ Elevated temperature ☐ High pressure ☐ Sonication ☐ Grinding ☐ Cutting ☐ Welding ☐ Other <i>(please specify)</i> :	☐ Gas release or oxygen deficiency? <i>If yes, include control measures.</i>	
Stage 4	☐ In contact with skin and eyes ☐ Inhalation ☐ Ingestion ☐ Direct entry <i>e.g. open wounds, injection</i> ☐ Other <i>(please specify)</i> :	☐ Weighing ☐ Spraying ☐ Pouring ☐ Addition of solid reagents ☐ Pipetting ☐ Filtering ☐ Shaking/mixing ☐ Centrifugation ☐ Use of sharps ☐ Elevated temperature ☐ High pressure ☐ Sonication ☐ Grinding ☐ Cutting ☐ Welding ☐ Other <i>(please specify)</i> :	☐ Other <i>(please specify)</i> :	

Persons who may be at risk:
Who is exposed to it?

- ☒ Departmental Staff
☐ Undergraduate students
☒ Postgraduate students
☐ Visitors
☐ Estates / cleaning / security staff
☐ Contractors, including service engineers
☐ Young workers
☐ Immuno-compromised individuals
☐ Pregnant women or women of reproductive age

☒ Others. Please specify:
 (Include names if appropriate)
 Volunteers that have signed up to the Birmingham River Champions project. They've had to confirm they've read a risk assessment (F-RAMP approved) within a sign up process

Notification method:
Groups at risk of exposure should be informed of any risk to which they may be exposed from this work activity

- ☐ Warning notice displayed on door
☐ Warning notice displayed within work area
 (e.g. laboratory bench, fume cupboard etc.)
☐ Hazards will be discussed with other workers
☐ Email notification to be sent to other workers

☐ Other. Please specify:

If nanomaterials are being used, please categorise the nanomaterial risk-level:

☐ HIGH:	Dry, dispersible nanomaterials or nanomaterial agglomerates / aggregates
☐ MEDIUM:	Nanomaterials suspended in liquids
☐ MEDIUM:	Nanomaterials attached to a 2D surface
☐ LOW:	Nanoparticles embedded into a solid 3D matrix

Approval for low- and medium-risk work involving nanomaterials is via principal investigator. Approval for high-risk nanomaterials work granted by either by the Safety Committee or jointly by the Safety Committee and the H&S Coordinator.

Control Measures

Could the use of hazardous substances be eliminated?
(If so, you should do so)

☐ YES ☒ NO

Could you substitute any of these materials for less hazardous alternatives or forms?
(If so, you should do so)

☐ YES ☒ NO

Controls and PPE Section 7, 8 & 15 of the Safety Data Sheet

What controls are required for these substances?

For procedures consisting of more than 4 stages, please continue in Appendix A.

	Engineering controls	Special requirements (e.g. special glassware; no vibration; Fluoro-plastic apparatus; no dry chemicals on heated surfaces; must be in the dark, not in spray/mist form, reduction in number of people exposed, authorised persons only)	PPE (PPE is a last resort and should be used in addition to all other control measures identified if the combination of these controls fails to achieve adequate control of exposure)
Stage 1	☒ None required ☐ A recirculating fume cupboard ☐ A ducted fume cupboard ☐ A glove box ☐ Local Exhaust Ventilation ☐ Other / additional control measures. Please specify: (e.g. a purpose-built facility, other specialist enclosure, well ventilated areas)	Polyethylene (PE) Gloves to be worn by all trained personnel undertaking the monitoring	☐ Eye protection? (State type required: e.g. safety spectacles, goggles, face shield) ☒ Gloves? (State type and thickness required, considering breakthrough times, permeation rates) ☐ Mask/RPE? (State type of mask/ filter) ☐ Laboratory coat or other clothing? (State type required) ☐ Other (please specify):

Stage 2	☒ None required ☐ A recirculating fume cupboard ☐ A ducted fume cupboard ☐ A glove box ☐ Local Exhaust Ventilation ☐ Other / additional control measures. Please specify: (e.g. a purpose-built facility, other specialist enclosure, well ventilated areas)	Polyethylene (PE) Gloves to be worn by all trained personnel undertaking the monitoring	☐ Eye protection? <i>(State type required: e.g. safety spectacles, goggles, face shield)</i> ☒ Gloves? <i>(State type and thickness required, considering breakthrough times, permeation rates)</i> ☐ Mask/RPE? <i>(State type of mask/ filter)</i> ☐ Laboratory coat or other clothing? <i>(State type required)</i> ☐ Other (please specify):
Stage 3	☐ None required ☐ A recirculating fume cupboard ☐ A ducted fume cupboard ☐ A glove box ☐ Local Exhaust Ventilation ☐ Other / additional control measures. Please specify: (e.g. a purpose-built facility, other specialist enclosure, well ventilated areas)		☐ Eye protection? <i>(State type required: e.g. safety spectacles, goggles, face shield)</i> ☐ Gloves? <i>(State type and thickness required, considering breakthrough times, permeation rates)</i> ☐ Mask/RPE? <i>(State type of mask/ filter)</i> ☐ Laboratory coat or other clothing? <i>(State type required)</i> ☐ Other (please specify):
Stage 4	☐ None required ☐ A recirculating fume cupboard ☐ A ducted fume cupboard ☐ A glove box ☐ Local Exhaust Ventilation ☐ Other / additional control measures. Please specify: (e.g. a purpose-built facility, other specialist enclosure, well ventilated areas)		☐ Eye protection? <i>(State type required: e.g. safety spectacles, goggles, face shield)</i> ☐ Gloves? <i>(State type and thickness required, considering breakthrough times, permeation rates)</i> ☐ Mask/RPE? <i>(State type of mask/ filter)</i> ☐ Laboratory coat or other clothing? <i>(State type required)</i> ☐ Other (please specify):

Monitoring Performance of control measures:

e.g. fume hood flow rates required, maintenance, examination and testing of engineering controls, PPE, RPE

☐ Fume cupboards and LEV:

- A thorough examination and test in the last 14 months, in accordance with University Policy and COSHH Regulations.
- A regular, routine maintenance schedule, in accordance with University Policy.
- User checks undertaken prior to each use, in accordance with University Policy.

☒ PPE/RPE:

- Users should meet the requirements for PPE and RPE, as set out in University Policy, including maintenance, inspection, and testing, and suitability assessment.
- Before each period of work, users must carry out an external visual inspection of all PPE to confirm suitability for the work activity and that it is not damaged or degraded so as to impair fit or performance.
- Respirators in regular use (other than one-shift disposable) must be thoroughly examined and, where appropriate, tested at least once a month or more frequently according to circumstances. Respirators used only occasionally should be examined and tested prior to next use and maintenance carried out as appropriate. N.B. The selection of a tight fit respirator for an individual wearer must include an appropriate fit test.

☐ Pressure vessels (e.g. liquid nitrogen transport dewars):

- Regular examination, inspection and testing, in accordance with University Policy and PSSR Regulations, and a written scheme is in place where required.
- Pressure vessels are checked regularly for general condition and leakage.

☐ Other measures not listed above (e.g. anemometers fitted to LEV equipment). Please specify:

Monitoring Workplace Exposure

Will monitoring for airborne contaminants or personal monitoring be required?

☐ YES

☒ NO

If yes, give details:

Monitoring Health Surveillance

Is there a known health impact from this level of exposure?

☐ YES

☒ NO

If yes, is health surveillance required to assess whether this health impact is developing in vulnerable individuals or all people exposed?

☐ YES

☒ NO

If yes, please raise a request with Occupational Health for Health Surveillance, via your line manager/PI by creating a new pre-placement at <https://birminghamuni.opasg2.com/>

Instruction and Training

Are any Standard Operating Procedures (SOPs) referred to in this protocol?

☐ YES

☒ NO

SOP reference number(s):

Specify training courses and/or special arrangements:

Storage and Waste Disposal

Storage

How should the materials be stored?

e.g. locked cupboard appropriately labelled and signed; stored away from other substances

Stored inside container away from non-trained individuals and wildlife.

Appropriate storage provision should be made prior to chemical purchase

Segregation *Section 10 of the Safety Data Sheet*

Is there any other chemical that these substances must not come in to contact with?

Procedure in case of reaction?

N/A – there are no particular risks of reactions with other substances under normal conditions

Disposal *Section 13 of the Safety Data Sheet*

How should the substance be disposed of?

Select all those that apply

☒ **Non-hazardous** *(Disposal via standard waste stream e.g. laboratory bin or sink)*

☐ **Halogenated solvent** - *an organic solvent containing halogens, i.e., Cl, F, Br, I (Disposal via specialist contractor through your local Health and Safety co-ordinator)*

☐ **Non-halogenated solvent** *(Disposal via specialist contractor through your local Health and Safety co-ordinator)*

☐ **Specialist hazardous waste disposal** - *including hazardous aqueous waste (Disposal via specialist contractor through your local Health and Safety co-ordinator)*

☐ **Treatment**

e.g. neutralisation of hazardous waste

Please detail:

Additional disposal details:

(Include 6-digit code and HP number if disposal is via external contractor and the coding is not undertaken centrally by your local H&S co-ordinator. See [Health and Safety Guidance - Hazardous Waste: Guidance on Assessment](#).)

Excess or expired reagent should be disposed of according to local regulations via the laboratory technician or manager.

Other Precautions and Emergency Procedures

Spillages or uncontrolled release *Section 6 of the Safety Data Sheet*

How should an accidental release or spillage be dealt with?

Consider both small- and large-scale spillages and release through failure of control measures

To control the hazards:

e.g. spread absorbent material on liquid spills, location of absorbent material, eliminate sources of ignition, isolation from power supply etc.

Prevent any further spillage if safe to do so.

The product must not come into contact with surface or ground water or enter the sewage system.

Put the leaked product into a suitable secondary container.

Dispose of all waste as hazardous materials.

To protect personnel:

Protection for personnel involved in the clean-up

Protective gloves to be worn throughout the full procedure.

To render site of emergency safe:

Equipment needed for decontamination, clean-up etc.

Plentiful supply of clean water.

Evacuation required? ☐

First Aid *Section 4 of the Safety Data Sheet*

What should be done in the case of:

Ingestion:

Rinse mouth out with plentiful supply of water. Get medical advice/attention immediately.

Eye contact:

Rinse with plentiful supply of water. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical advice/attention immediately if symptoms appear.

Inhalation:

Ensure surveyors are outdoors / getting fresh air away from the accident scene. First aid trained surveyors could administer artificial respiration. Get medical advice/attention immediately if symptoms appear.

Skin contact:

Take off immediately all contaminated clothing. Rinse skin with plentiful supply of water. Get medical advice/attention immediately if symptoms appear

In the event of contamination or exposure, it is important to be able to contact a first aider without delay. Ensure you are aware of your local first aiders prior to commencement of work.

Fire Precautions *Section 5 of the Safety Data Sheet*

What actions should be taken in the event of fires involving these substances?

Cut off ignition sources and extinct by a suitable media.

The first step should be to raise the alarm BUT if a fire extinguisher is needed, what extinguisher(s) should be used?

e.g. CO₂, foam, fire blanket, Class D Powder extinguisher

Water (mist), powder, carbon dioxide and dry sand.

Only tackle a fire if you have received training and it is safe to do so. Never put yourself at risk.

Emergency contact:

Name(s) and contact telephone number(s)

Review

Review date:

Maximum interval 3 years. Please note, some areas (including Chemistry) set a maximum review date of 1 year. Please check locally if unsure.

18/09/2029

You should review your risk assessment if you think it may no longer be valid (e.g. following significant changes to hazards, such as a change in process, materials, quantities, or following an accident, incident or ill-health.)

Assessment of Risk

Is the level of risk acceptable (are hazards to health adequately controlled with control measures in place)?

☒ YES

☐ NO

Overall assessment of risk:

Taking into account the above factors, what level of risk does the procedure pose?

☐ EXTREME: likely to present significant or permanent health effects to the majority of individuals

☐ HIGH: likely to present a significant risk to health to the majority of individuals

☐ MEDIUM: may present a significant risk to health to a minority of individuals

☒ LOW: the majority of individuals will be unaffected with only a minimal risk to health

☐ INSIGNIFICANT: no foreseeable risk of injury

N.B. Where risks associated with dangerous substances and/or explosive atmospheres are not adequately controlled, a DSEAR assessment should be undertaken to identify and assess the risks and determine any additional control measures required.

Declaration

Note: If working with nanomaterials, approval for low- and medium-risk work is via the Principal Investigator. Approval for high-risk nanomaterials work is to be granted by either by the Safety Committee or jointly by the Safety Committee and the Health and Safety Coordinator.

Assessor(s) Name:	Assessor(s) Signature:	Date:
Caroline Roper	Caroline Roper	18/09/2026

The line manager, supervisor or principal investigator should sign below to show that the assessment is a correct and reasonable reflection of the hazards and of the control measures and actions required:

Supervisor(s) Name:	Supervisor(s) Signature:	Date:

Appendix

Appendix 1 CLP Toxicity rating

Toxicity Rating	H number and statement
1/2	H300 Fatal if swallowed H310 Fatal in contact with skin H330 Fatal if inhaled

3	H301 Toxic if swallowed H311 Toxic in contact with skin H331 Toxic if inhaled
3 (ASPT1)	H304 May be fatal if swallowed and enters airways
4	H302 Harmful if swallowed H312 Harmful in contact with skin H332 Harmful if inhaled
Not classified as toxic	